

Light #2

1. A *diverging* lens has a focal length of 9.0 cm, and an object is placed 3.0 cm from the lens.
 - (a) What would be the distance of the image from the lens?

 - (b) What is the magnification of the image?

 - (c) Will the image be real or virtual? How do you know?

2. A rutabaga, which has a height of 44 cm is placed 10 cm in front of a lens. The image produced has a height of 66 cm and is *inverted*.
 - (a) What would be the distance of the image from the lens?

 - (b) What is the magnification of the image? (*Be careful with signs!*)

 - (c) What is the focal length of the lens?